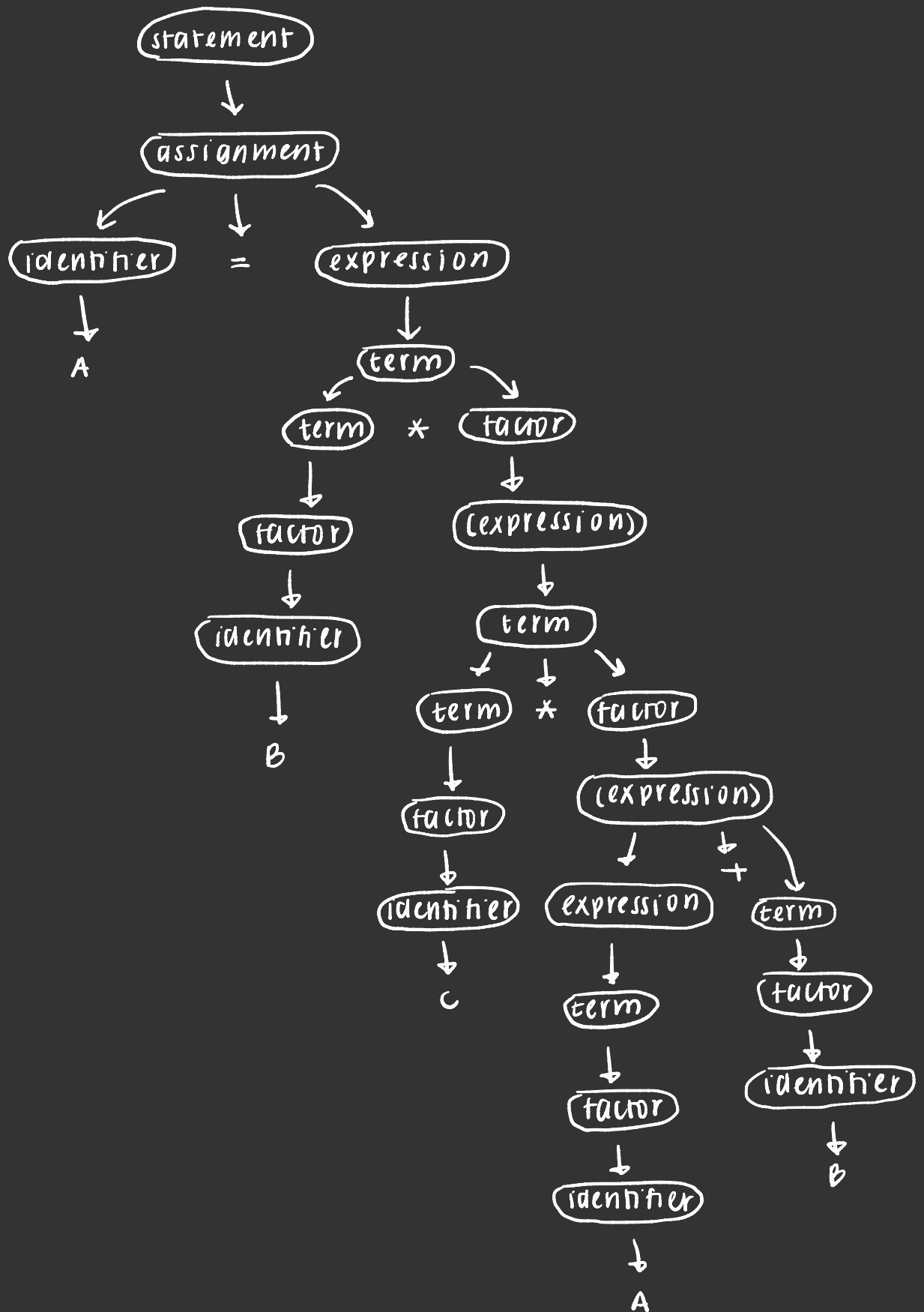




1. Parse Tree and Leftmost Derivation

$$A = B * (C * (A + B))$$

Statement

$\Rightarrow$ Assignment
 $\Rightarrow$ Identifier = Expression
 $\Rightarrow A = \text{Expression}$
 $\Rightarrow A = \text{Term}$
 $\Rightarrow A = \text{Term} * \text{Factor}$
 $\Rightarrow A = \text{Factor} * \text{Factor}$
 $\Rightarrow A = \text{Identifier} * \text{Factor}$
 $\Rightarrow A = B * \text{Factor}$
 $\Rightarrow A = B * (\text{Expression})$
 $\Rightarrow A = B * (\text{Term})$
 $\Rightarrow A = B * (\text{Term} * \text{Factor})$
 $\Rightarrow A = B * (\text{Factor} * \text{Factor})$
 $\Rightarrow A = B * (\text{Identifier} * \text{Factor})$
 $\Rightarrow A = B * (C * \text{Factor})$
 $\Rightarrow A = B * (C * (\text{Expression}))$
 $\Rightarrow A = B * (C * (\text{Expression} + \text{Term}))$
 $\Rightarrow A = B * (C * (\text{Term} + \text{Term}))$
 $\Rightarrow A = B * (C * (\text{Factor} + \text{Term}))$
 $\Rightarrow A = B * (C * (\text{Identifier} + \text{Term}))$
 $\Rightarrow A = B * (C * (A + \text{Term}))$
 $\Rightarrow A = B * (C * (A + \text{Factor}))$
 $\Rightarrow A = B * (C * (A + \text{Identifier}))$
 $\Rightarrow A = B * (C * (A + B))$

2. Scope Concepts

a. Static scoping results

$u = 180$

add(v)	<u>Global</u> $u = 69 + 42 + 69 = 180$ $v = 69$ $w = 17$ $z = 69$
bar(add)	<u>Global</u> $u = 42$ $v = 69$ $w = 17$ <u>Local</u> $u = 17$
foo(u, 13)	<u>Global</u> $u = 42$ $v = 69$ $w = 17$ <u>Local</u> $v = 17$
main	<u>Global</u> $u = 42$ $v = 69$ $w = 17$

b. Dynamic Scoping with Deep Binding

$u = 126$

add()	<u>Non-local</u> $v = 42$ $x = 42$ $w = 13$ Local $z = 42$ $u = 42 + 42 + 42 = 126$
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bar()	<u>Non-local</u> v = 42 x = 42 w = 13 <u>Local</u> u = 13
foo(u, 13)	<u>Local</u> v = 42 x = 42 w = 13
main	<u>Global</u> u = 42 v = 69 w = 17

c. Dynamic Scoping with Shallow Binding

u = 92

fun(v)	<u>Non-local</u> u = 13 <u>Global</u> u = 42+13+42 = 97
bar()	<u>Non-local</u> x = 42 v = 42 w = 13 <u>Local</u> u = 13
foo(u, 13)	<u>Local</u> x = 42 v = 42 w = 13
main	<u>Global</u> u = 42 v = 69 w = 17